

Host genetic background modulates *ROOMIE1*-associated microbial recruitment and transcriptomic responses in *Lotus japonicus*

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Background

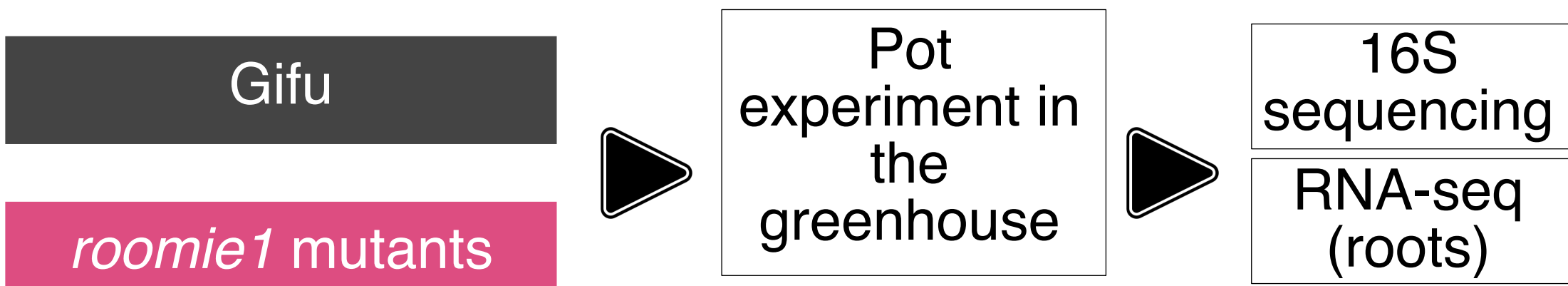
- We previously identified *ROOMIE1* is a key host gene that exerts genetic control over the establishment and composition of the rhizosphere microbiome.
- Natural *ROOMIE1* haplotype variation was shown to drive significant microbiome differences among closely related plant accessions, demonstrating its role in maintaining fine-scale ecological diversity.
- *ROOMIE1* is annotated as an ARM-repeat superfamily protein. *ROOMIE1* and its putative Arabidopsis ortholog AT1G27210 both contain a HEAT domain, a protein-protein interaction motif.

Objectives

- We investigated the mechanistic basis of *ROOMIE1*'s control using transcriptomic profiling of *roomie1* mutants and natural accessions from two distinct locations in Japan that carry allelic variation within this gene.
- We performed rhizospheric microbiome profiling of the mutants and the natural accessions to detect the links between *ROOMIE1*-associated microbial recruitment and transcriptomic responses.

Methods

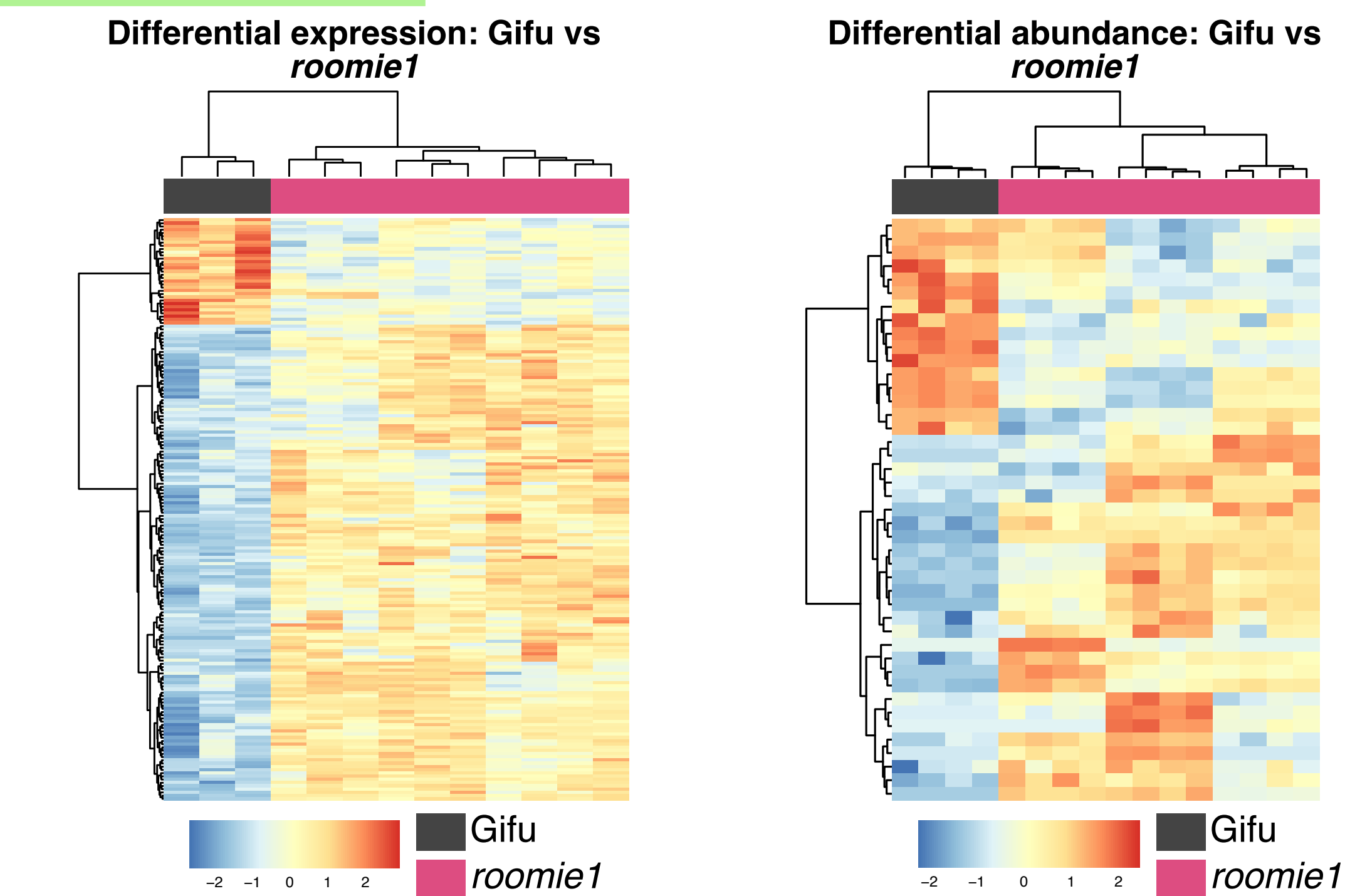
1. Comparing Gifu wild-type against *LORE1* insertion mutants



2. Population transcriptomics & microbe-gene network



Results



GO enrichment analysis of the genes up-regulated in the mutants

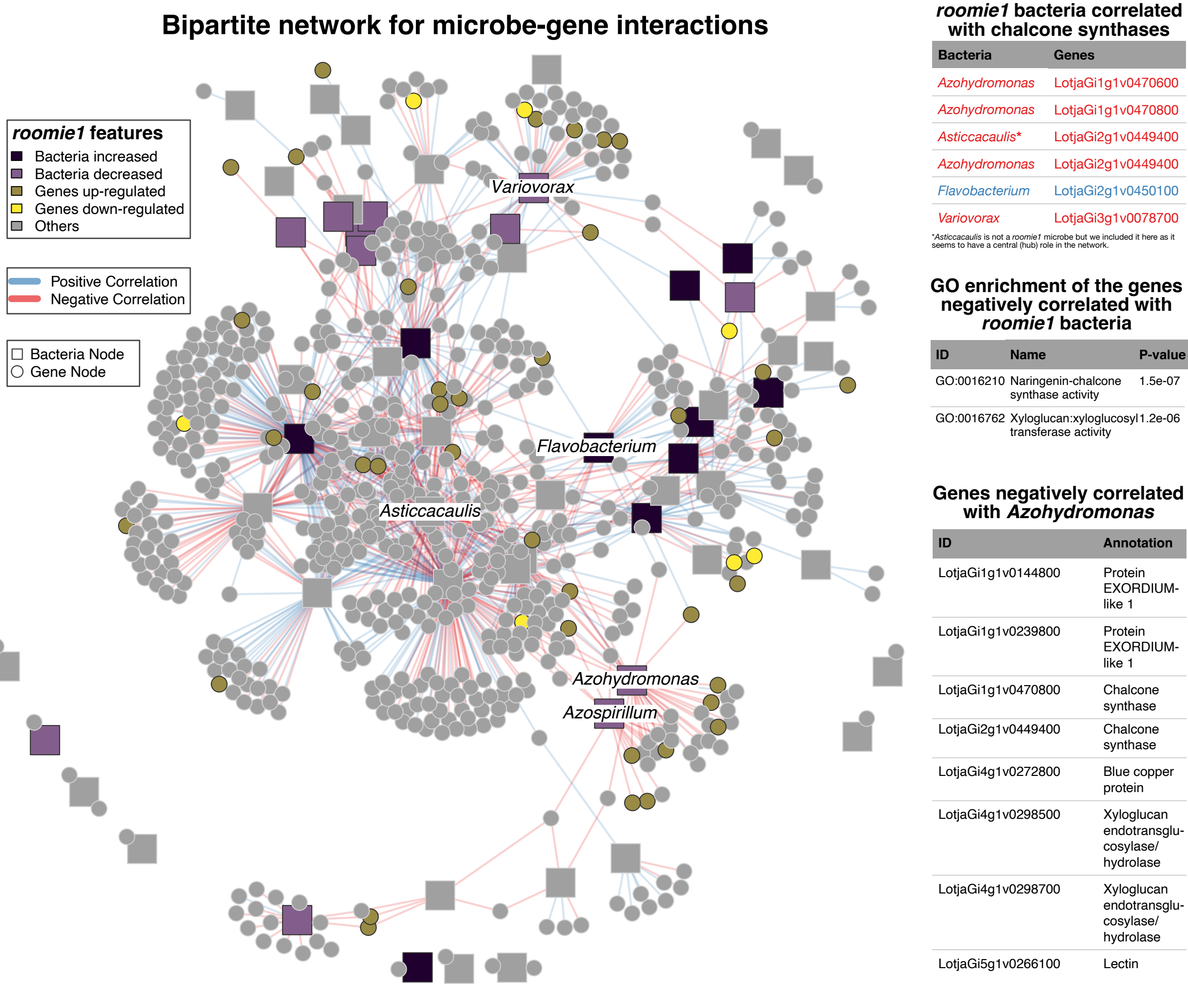
ID	Name	Annotated	Significant	Expected	P-value
GO:0016210	Naringenin-chalcone synthase activity	18	5	00.11	2.2e-09
GO:0008171	O-methyltransferase activity	49	4	00.03	1.7e-05
GO:0016782	Xyloglucan:xyloglucosyl transferase activity	29	3	00.18	1.0e-04

Bacteria decreased in the mutants

Bacteria	Elusimicrobium	Anaerobacterium	Nocardioides
Adhaeribacter	Elusimicrobium	Anaerobacterium	Nocardioides
Arthrobacter	Arthrobacter	Anaerobacterium	Nocardioides
Azohydromonas	Azohydromonas	Anaerobacterium	Nocardioides
Massilia	Massilia	Anaerobacterium	Nocardioides
Mitsurina	Mitsurina	Anaerobacterium	Nocardioides
Paenibacillus	Paenibacillus	Anaerobacterium	Nocardioides
Pseudomonas	Pseudomonas	Anaerobacterium	Nocardioides
Streptomyces	Streptomyces	Anaerobacterium	Nocardioides
Urethrobacter	Urethrobacter	Anaerobacterium	Nocardioides
Variovorax	Variovorax	Anaerobacterium	Nocardioides

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Perspectives

- With WT vs. *roomie1* mutants analysis and haplotype-specific networks from natural plants, we have been accumulating key bacteria and genes that can explain *ROOMIE1*'s role in microbiome establishment.
- We aim to confirm these candidate features via mutant experiments, microbial isolation, and metabolomic studies.

Acknowledgements

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